

Assignment 1: AI System Specification

Course: Introduction to Artificial Intelligence

System Selected: E: A Smart Traffic Signal Controller for reducing congestion in cities.

Task 1: PEAS Specification

- **Performance Measure:**
 - **Average Wait Time:** Minimizing the time vehicles spend waiting at a red light.
 - **Intersection Throughput:** Maximizing the total number of vehicles that successfully cross the intersection per hour.
 - **Emergency Vehicle Delay:** Reducing the wait time for emergency vehicles (like ambulances and fire trucks) to zero.
 - **Emission Levels:** Reducing fuel wastage and carbon emissions caused by idling vehicles.
- **Environment:** Busy urban intersections with mixed traffic (cars, buses, bikes), pedestrians, varying weather conditions (rain, fog), and unexpected events like accidents or road construction.
- **Actuators:** Traffic light indicators (Red, Yellow, Green LED signals) and digital road signs/display boards to give instructions to drivers.
- **Sensors:** High-resolution RGB cameras mounted on poles, inductive loop detectors embedded under the road surface, and wireless transponder receivers to detect approaching emergency vehicles.

Task 2: Environment Classification

- **Partially Observable:** The controller can only scan and see its own intersection; it cannot see the traffic status of the entire city.
- **Multi-agent:** Multiple smart signal controllers at neighboring intersections operate at the same time and must coordinate with each other.
- **Stochastic:** Traffic flow is highly unpredictable because accidents, sudden vehicle breakdowns, or human driving behavior cannot be calculated with 100% certainty.
- **Sequential:** A decision made to extend a green light right now directly affects the accumulation and flow of traffic in the next few minutes.
- **Dynamic:** The traffic situation changes continuously while the AI system is computing and deciding whether to change the lights.
- **Continuous:** Factors like vehicle speeds, exact physical locations on the road, and time durations are measured in continuous numbers, not fixed steps.

Task 3: Critical Analysis

1. Greatest Technical Challenge

The greatest technical challenge for an AI engineer building this system is dealing with the **Multi-agent** and **Dynamic** nature of the environment. If one signal works independently and clears its traffic by keeping the light green for too long, it might suddenly flood the next intersection with cars, creating a massive traffic jam there. Designing an AI that instantly reacts to changing traffic while collaborating perfectly with nearby signals is highly complex.

2. Utility Function

A suitable utility function to balance traffic flow and pedestrian safety can be defined as:

$$U = w_1 \cdot (\text{Throughput}) - w_2 \cdot (\text{Pedestrian Wait Time})$$

(Where w_1 and w_2 represent the weights or importance given to each factor).

Behavior Change: If the weight of the pedestrian wait time (w_2) is doubled, the AI agent will prioritize human safety and convenience over vehicle speed. The system will trigger pedestrian crossing lights much more frequently. While this might slightly slow down the overall car traffic, it will make the streets significantly safer and friendlier for walking citizens.

=> Structured Representation:

JSON

```
{
  "system_name": "Smart Traffic Signal Controller",
  "peas": {
    "performance_measures": [
      "average_vehicle_wait_time",
      "intersection_throughput",
      "emergency_vehicle_delay",
      "carbon_emission_levels"
    ],
    "environment": "Busy urban intersections with unpredictable
traffic and weather",
    "actuators": [
      "traffic_light_relays",
      "digital_message_signs"
    ],
    "sensors": [
      "high_resolution_cameras",
      "inductive_loop_detectors",
      "emergency_vehicle_receivers"
    ]
  },
  "environment_classification": {
    "observability": {
      "choice": "Partially Observable",
      "justification": "The controller can only monitor its immediate
intersection, not the entire city's traffic state."
    },
    "agents": {
      "choice": "Multi-agent",
      "justification": "Multiple smart signals at adjacent
intersections must interact and coordinate with each other."
    },
    "determinism": {
      "choice": "Stochastic",
      "justification": "Traffic volume changes randomly due to
accidents, vehicle breakdowns, or human behavior."
    }
  }
}
```

```
    },
    "episodicity": {
        "choice": "Sequential",
        "justification": "Current timing decisions directly impact
traffic accumulation and flow in subsequent time intervals."
    },
    "dynamism": {
        "choice": "Dynamic",
        "justification": "The traffic situation continuously changes
while the agent is deciding on the next signal phase."
    },
    "continuity": {
        "choice": "Continuous",
        "justification": "Variables like time duration, vehicle speed,
and exact spatial positions are continuous."
    }
},
"utility_function": "U = 0.7 * Throughput - 0.3 *
Pedestrian_Wait_Time"
}
```